计算机硬件系统设计

多周期MIPS CPU ---2

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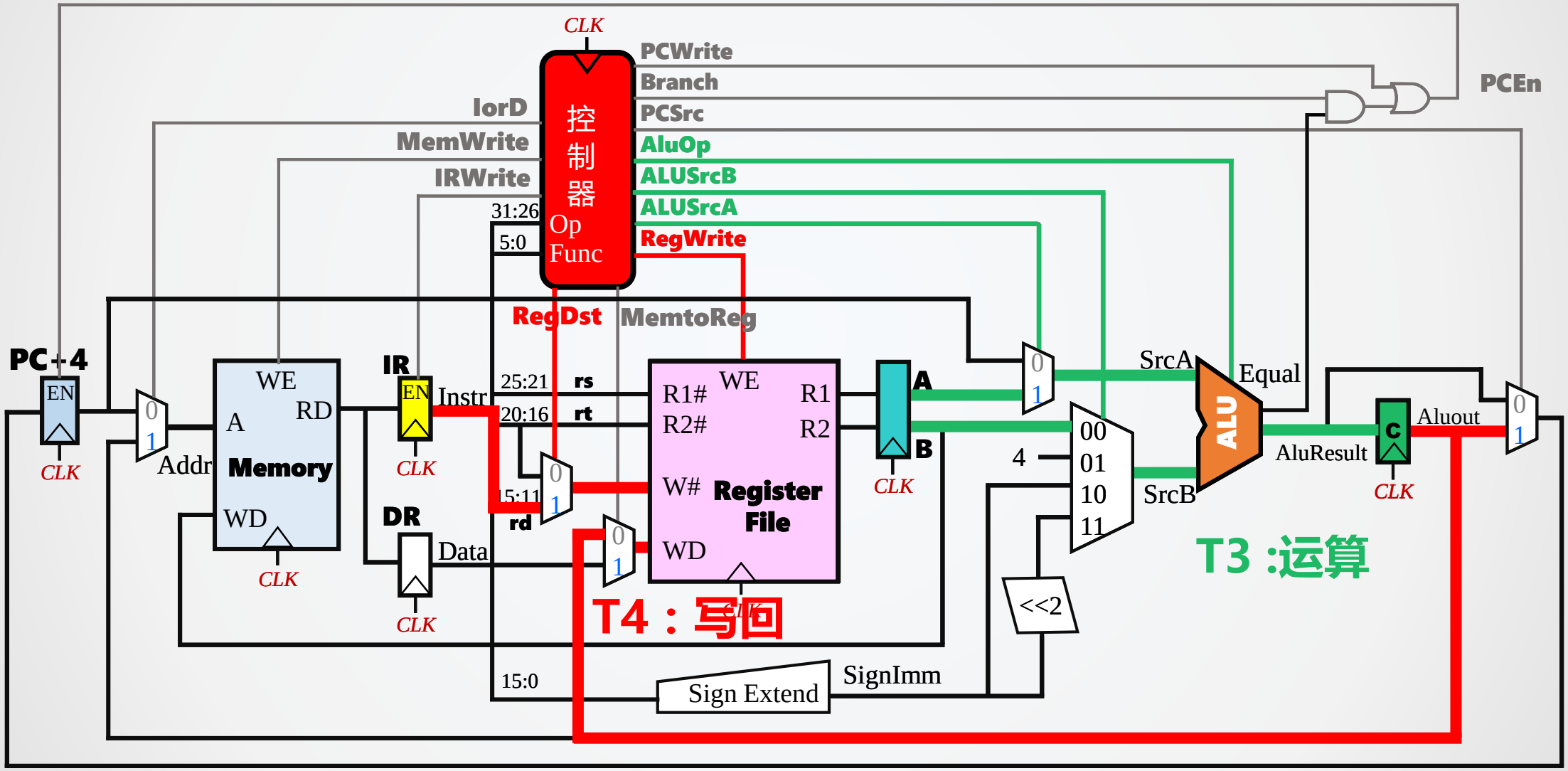
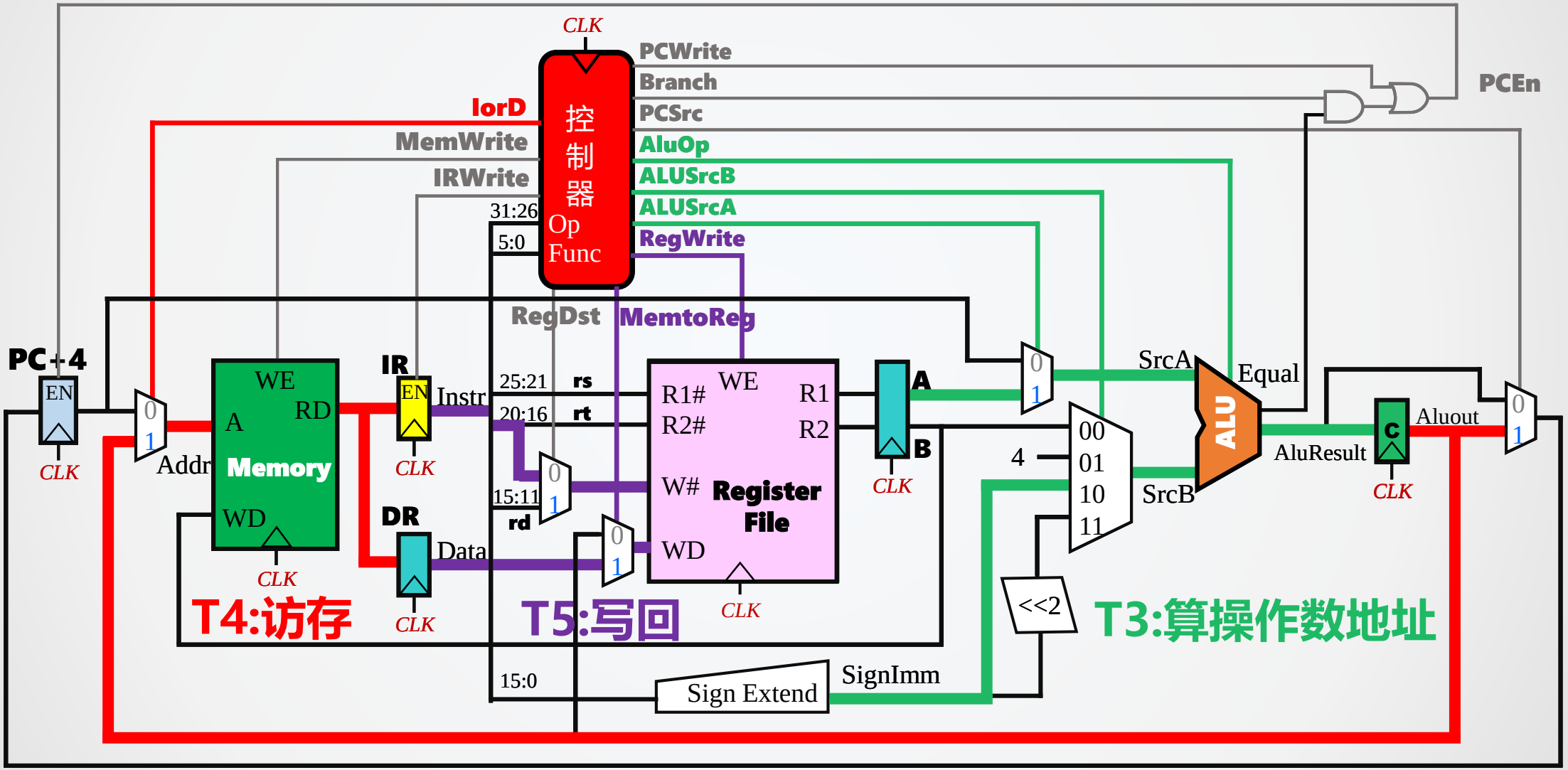
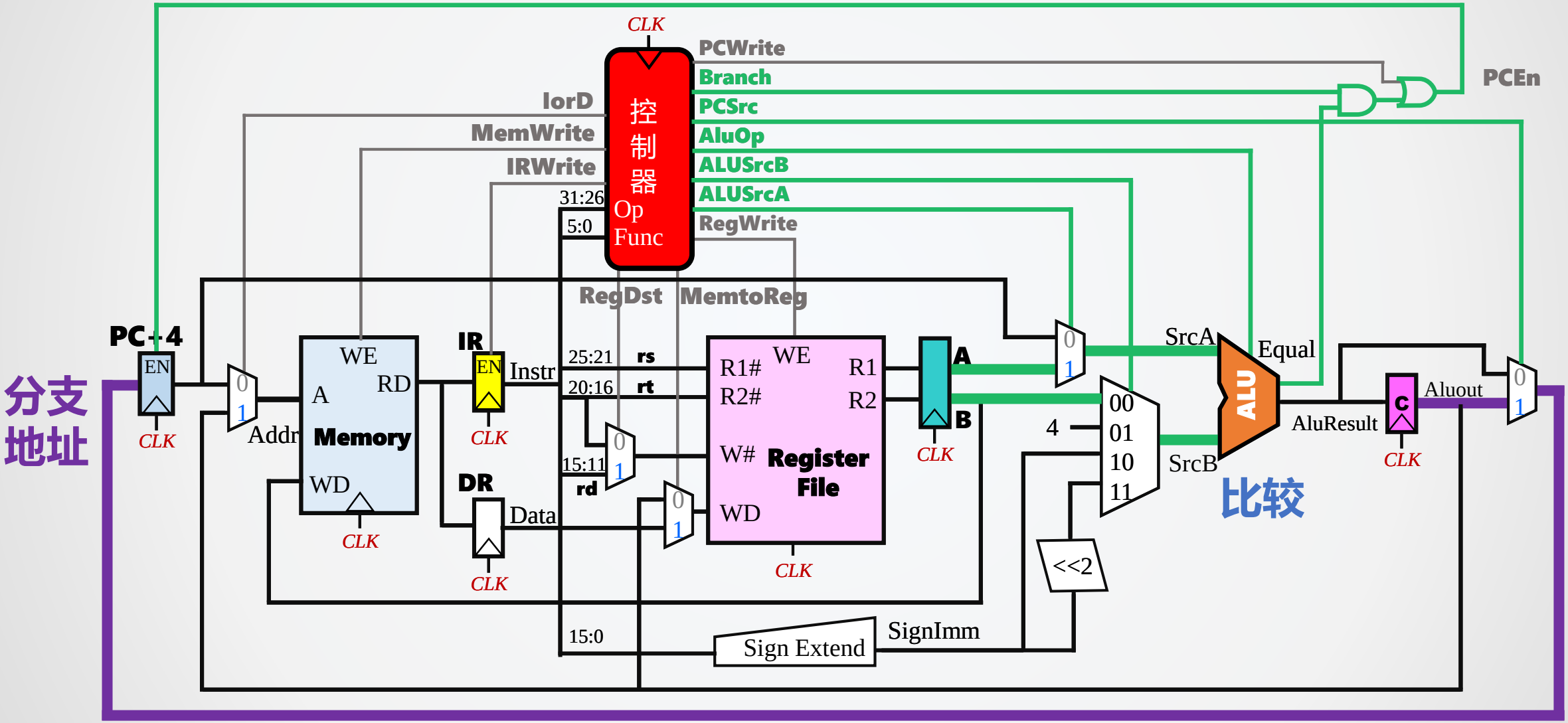
R型指令执行状态周期T3~T4



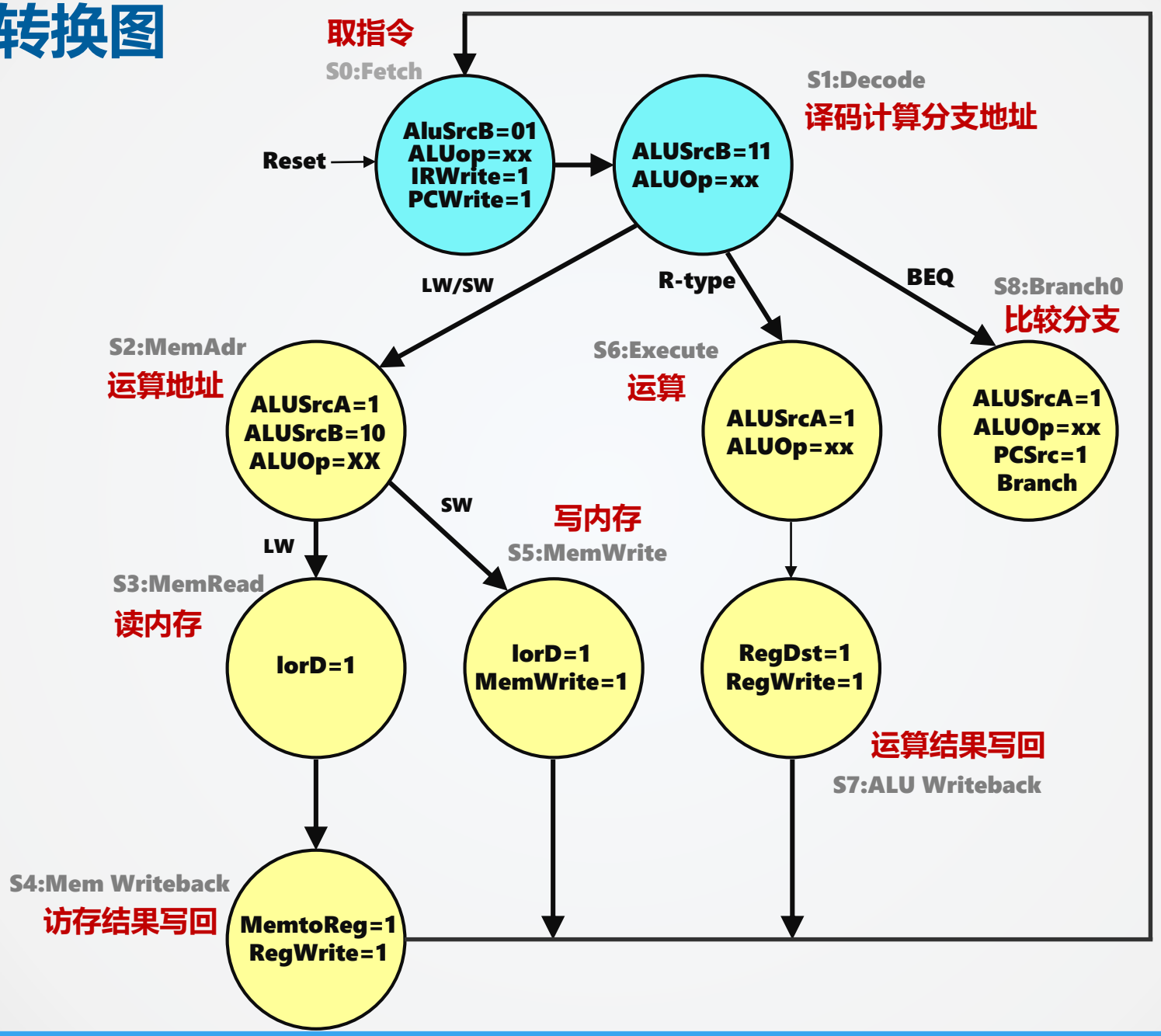
LW指令执行状态周期T3~T5



Beq指令执行状态周期T3



多周期状态转换图


取指令
S0:Fetch
AluSrcB=01
ALUop=xx
IRWrite=1
PCWrite=1
Reset
S1:Decode
译码计算分支地址
ALUSrcB=11
ALUOp=xx
LW/SW
R-type
BEQ
S8:Branch0
比较分支
ALUSrcA=1
ALUOp=xx
PCSrc=1
Branch
S2:MemAdr
运算地址
ALUSrcA=1
ALUSrcB=10
ALUOp=XX
S6:Execute
运算
ALUSrcA=1
ALUOp=xx
SW
写内存
S5:MemWrite
lorD=1
MemWrite=1
LW
S3:MemRead
读内存
lorD=1
RegDst=1
RegWrite=1
运算结果写回
S7:ALU Writeback
S4:Mem Writeback
访存结果写回
MemtoReg=1
RegWrite=1

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下节课再见...

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